Sina Sharifi

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Professional Summary

Ph.D. candidate in Electrical and Computer Engineering at Johns Hopkins University with experience in developing machine learning algorithms in PyTorch. Research spans diffusion models and flow matching, optimization, out-of-distribution detection, adversarial robustness, and neural network verification. First-author publications at ICML, ECCV, TMLR, and Automatica.

Education

Johns Hopkins University — Baltimore, MD

- Ph.D. Candidate in Electrical and Computer Engineering — 2022–Present
  Advisor: Prof. Mahyar Fazlyab (*Expected Graduation December 2027*)
- M.S.E. in Electrical and Computer Engineering — 2022–2024

University of Tehran — Tehran, Iran

- B.Sc. in Electrical Engineering — 2016–2021
- Minor Degree in Computer Engineering

Research Experience

Graduate Research Assistant, Johns Hopkins University — Jan 2022–Present

- *Generative Modeling and Out-of-Distribution Detection*
  - Developed few-shot guidance methods for steering pretrained diffusion and flow-matching models using as few as 10 target examples.
  - Designed a flow-matching-based OOD detection framework using trajectory-derived statistics and evaluated it across diverse near- and far-OOD settings, including a 17% improvement on ImageNet-scale experiments.
  - Built reproducible PyTorch and Slurm pipelines for multi-seed GPU experiments, hyperparameter sweeps, ablations, and large-scale model and sample evaluation.
- *Sampling-Based Optimization and Control*
  - Established a connection between model predictive path integral control (MPPI) and preconditioned gradient descent, enabling convergence and finite-sample analysis of sampling-based control algorithms.
- *Optimization and Learning Algorithms*
  - Developed safe-gradient-flow algorithms for nonconvex bilevel optimization, using control barrier functions and adaptive line-search procedures to maintain feasibility.
  - Proved convergence guarantees and validated the methods on machine learning and optimization benchmarks, leading to publications in TMLR, IEEE L-CSS, and ACC.
- *Neural Network Verification and Robustness*
  - Developed curvature- and Lipschitz-based bounds for neural networks and used them in verification frameworks for certifying neural autonomous systems, leading to publications in Automatica and ICRA.
  - Developed Taylor-expansion-based adversarial training methods and end-to-end PyTorch pipelines for robustness evaluation and ablation studies, leading to a publication at ICML.

AI/ML Engineer Intern, Akoncagua AI — Aug 2026–Nov 2026

• Building end-to-end machine learning systems for proprietary, domain-specific applications, including data preprocessing, model development, evaluation, and deployment, and developing diffusion-based generative models for domain-specific problems.

• Working with large language models, including tokenization, parameter-efficient fine-tuning, and distributed training and inference on multi-GPU systems using AWS EC2.

Machine Learning Intern, Mobile Communications Laboratory, University of Tehran — Aug 2020–Mar 2021

• Developed RNN- and LSTM-based models for bandwidth prediction from communication-network time-series data, including feature engineering, preprocessing, training, and evaluation.

Technical Skills

Programming: Python, C++, MATLAB, SQL

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, NumPy, SciPy, CVXPY

Generative AI & ML Systems: Unsloth, Ollama, Multi-GPU Training, CUDA, Slurm, Weights & Biases (W&B)

Cloud Platforms & Infrastructure: AWS (EC2, S3), Google Cloud (GCP), Docker, Git

Selected Publications (full list on Google Scholar)

[1] M. Fazlyab, S. **Sharifi**, and J. Wang, "Model predictive path integral control as preconditioned gradient descent," *IEEE Control Systems Letters (**L-CSS**)*, 2026.

[2] S. **Sharifi**, N. Abolfazli, M. Fazlyab, and E. Y. Hamedani, "Perturbed gradient descent via convex quadratic approximation for nonconvex bilevel optimization," *Transactions on Machine Learning Research (**TMLR**)*, 2026.

[3] S. **Sharifi** and M. Fazlyab, "Provable bounds on the hessian of neural networks: Derivative-preserving reachability analysis," *To be published at Automatica*, 2026.

[4] S. **Sharifi**, B. Safaei, and M. Fazlyab, "Flow Matching for Out-of-Distribution Detection," Under review, 2026.

[5] S. **Sharifi**, E. Y. Hamedani, and M. Fazlyab, "Sequential QCQP for bilevel optimization with line search," *IEEE Control Systems Letters (**L-CSS**)*, 2025.

[6] S. **Sharifi**, T. Entesari, and M. Fazlyab, "Compositional curvature bounds for deep neural networks," in *International conference on machine learning (**ICML**)*, 2024.

[7] S. **Sharifi**, T. Entesari, B. Safaei, V. Patel, and M. Fazlyab, "Gradient-regularized out of distribution detection," in *European Conference on Computer Vision (**ECCV**)*, 2024.

[8] T. Entesari, S. **Sharifi**, and M. Fazlyab, "Reachlipbnb: A branch-and-bound method for reachability analysis of neural autonomous systems using lipschitz bounds," in *IEEE International Conference on Robotics and Automation (**ICRA**)*, 2023.

Honors and Awards

- Mathematical Institute for Data Science (**MINDS**) Fellowship, Johns Hopkins University 2024
- ECE Departmental Fellowship, Johns Hopkins University 2022
- Ranked 11th among 112 Electrical Engineering B.Sc. students, University of Tehran 2021

Professional Services

• **Graduate Iranian Organization, Johns Hopkins University** Jun 2024–Feb 2026
Organized community, cultural, and social events for Iranian graduate students.

• **Whiting Internships in Science and Engineering (WISE)** Jan 2026–May 2026
Mentored undergraduate students pursuing research and internship opportunities.

• **Peer Reviewer** 2025–2026
Reviewed submissions for leading conferences and journals, including **CVPR**, **NeurIPS**, **ECCV**, IEEE Transactions on Automatic Control (**TAC**), **CDC**, and IEEE **L-CSS**.

• **Teaching and Course Support, Johns Hopkins University** 2023–2025
Head Course Assistant for Control Systems; Teaching Assistant for Modern Convex Optimization and Introduction to Nonlinear Systems.